

# Linux Network Emulation and SDN Lab Using Mininet

## Overview

Designed and tested a virtual software-defined networking (SDN) environment using the Mininet VM and MiniEdit. Configured hosts, switches, and router interfaces, analyzed routing behavior, tested inter-network communication, monitored TCP/UDP sockets, and implemented Linux firewall rules using UFW.

## Environment & Tools

- Platform: Mininet VM
- Network Emulator: Mininet
- Topology Editor: MiniEdit
- Operating System: Linux
- Networking Utilities: `ifconfig`, `route`, `ping`, `ss`, `ps`
- Traffic Testing Tool: iPerf
- Firewall Management: UFW (Uncomplicated Firewall)

## Network Topology

Topology created and tested:



## Device Roles

- h1 / h2: End hosts
- s1: Layer 2 switch
- r2: Router connecting two subnets

## Technical Tasks Performed

### 1. Interface and MAC Address Analysis

Identified router interfaces and hardware addresses using Linux networking commands.

| Router Interfaces |             |
|-------------------|-------------|
| Interface         | MAC Address |

|         |                   |
|---------|-------------------|
| r2-eth0 | d6:c2:2a:70:cc:7f |
| r2-eth1 | 32:26:63:6b:41:16 |

### Skills Demonstrated

- Linux network interface analysis
- MAC address identification
- CLI-based troubleshooting

## 2. Routing Table Configuration & Analysis

Examined routing behavior and interpreted CIDR-based routing entries.

| Router Table        |      |                    |
|---------------------|------|--------------------|
| Destination Network | CIDR | Outgoing Interface |
| 10.0.0.0            | /8   | r2-eth0            |
| 11.0.0.0            | /8   | r2-eth1            |

### Concepts Applied

- Static routing
- CIDR notation
- Subnet analysis
- Route interpretation

## 3. Network Connectivity Testing

Used ICMP ping testing to verify communication between hosts and router interfaces.

### Findings

- Initial communication between h2 and 11.0.0.10 failed due to missing routing configuration.

- After reconfiguring h2 into the 11.0.0.0/8 subnet, communication succeeded.
- Verified successful routing between separate subnets through r2.

### **Skills Demonstrated**

- Network troubleshooting
- Layer 3 routing analysis
- Gateway configuration
- ICMP diagnostics

## **4. Default Gateway & Inter-Subnet Routing**

Configured default gateways to enable communication between separate networks.

### **Commands Used**

- route add default gw 10.0.0.10
- route add default gw 11.0.0.10

### **Outcome**

Successfully enabled communication between:

- 10.0.0.0/8
- 11.0.0.0/8

### **Concepts Demonstrated**

- Default gateway configuration
- Router-based packet forwarding
- Inter-network communication

## **5. Switch Behavior Analysis**

Verified Layer 2 switch functionality within the topology.

### **Key Observation**

Confirmed that the switch **s1** could not be pinged directly because:

- Layer 2 switches do not use IP addresses by default
- Ping requires Layer 3 IP communication

## Networking Concepts

- OSI model understanding
- Layer 2 vs Layer 3 communication
- Ethernet switching fundamentals

## 6. TCP & UDP Socket Analysis

Analyzed active network sockets using the Linux **ss** utility.

### TCP Socket States Observed

- LISTEN
- ESTABLISHED
- TIME-WAIT

### UDP Socket States Observed

- UNCONN

### Skills Demonstrated

- Socket inspection
- TCP state analysis
- UDP behavior analysis
- Linux network diagnostics

## 7. Process & Port Investigation

Mapped active sockets to running processes using Linux process management tools.

|                     |
|---------------------|
| Identified Services |
|---------------------|

| Protocol | Port | Process | PID  |
|----------|------|---------|------|
| UDP      | 5001 | iperf   | 3140 |
| TCP      | 5001 | iperf   | 3137 |

### Commands Used

- ss -a
- ps u -p <PID>

### Skills Demonstrated

- Linux process analysis
- Service identification
- Port-to-process mapping

## 8. Firewall Configuration & Testing

Configured Linux firewall rules using UFW and verified access control behavior.

### Firewall Actions

- Enabled firewall protection: ufw enable
- Allowed iPerf TCP traffic: ufw allow 5001/tcp

### Observations

- Initial iPerf connection attempts failed due to blocked incoming traffic.
- After adding firewall rules, TCP connectivity succeeded.

### Skills Demonstrated

- Linux firewall management
- Access control configuration
- Network security testing
- Service rule implementation

## Key Networking Concepts Applied

- TCP/IP Networking
- CIDR Subnetting
- Static Routing
- Default Gateways
- Layer 2 Switching
- Layer 3 Routing
- TCP/UDP Socket States
- Linux Firewall Administration
- Network Troubleshooting
- Client-Server Communication

## Appendix / Technical Evidence

### Appendix A: Routing Table Verification

```

Host: r2@mininet-vm
RX packets 0 bytes 0 (0.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

r2-eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
ether 32:26:63:6b:41:16 txqueuelen 1000 (Ethernet)
RX packets 0 bytes 0 (0.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vm:~# ^C
root@mininet-vm:~# ifconfig r2-eth0 10.0.0.10 netmask 255.0.0.0 up
root@mininet-vm:~# ifconfig r2-eth1 11.0.0.10 netmask 255.0.0.0 up
root@mininet-vm:~# route -n
Kernel IP routing table
Destination Gateway Genmask Flags Metric Ref
Use Iface
10.0.0.0 0.0.0.0 255.0.0.0 U 0 0
0 r2-eth0
11.0.0.0 0.0.0.0 255.0.0.0 U 0 0
0 r2-eth1
root@mininet-vm:~# ^C
root@mininet-vm:~# █

```

**Figure 1:** Routing table on **r2** showing routes for **10.0.0.0/8** and **11.0.0.0/8**

### Appendix B: Successful Inter-Subnet Ping

```

"Host: h2"@mininet-vm
h2-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 150
    inet 10.0.0.2 netmask 255.0.0.0 broadcast 10.255.2
    ether 76:91:f0:4c:57:c1 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collis

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 641 bytes 107256 (107.2 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 641 bytes 107256 (107.2 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collis

root@mininet-vm:~# ping -c 3 11.0.0.10
connect: Network is unreachable
root@mininet-vm:~# ^C
root@mininet-vm:~# ^C
root@mininet-vm:~# ifconfig h2-eth0 11.0.0.1 netmask 255.0.0.0 up
root@mininet-vm:~# ping -c 3 11.0.0.10
PING 11.0.0.10 (11.0.0.10) 56(84) bytes of data:
64 bytes from 11.0.0.10: icmp_seq=1 ttl=64 time=0.126 ms
64 bytes from 11.0.0.10: icmp_seq=2 ttl=64 time=0.034 ms
64 bytes from 11.0.0.10: icmp_seq=3 ttl=64 time=0.035 ms

--- 11.0.0.10 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2039ms
rtt min/avg/max/mdev = 0.034/0.065/0.126/0.043 ms
root@mininet-vm:~# █

```

**Figure 2:** ICMP communication between hosts after default gateway configuration

## Appendix C: Firewall Rule Configuration

```

"Host: h1"@mininet-vm
allow|deny|reject|limit [in|out [on INTERFACE]] [log|log-all] [proto
PROTOCOL] [from ADDRESS [port PORT | app APPNAME ]] [to ADDRESS [port
PORT | app APPNAME ]] [comment COMMENT]
root@mininet-vm:~# ufw allow 5001/tcp
Rule added
root@mininet-vm:~# ufw status
Status: active

To Action From
--
5001/tcp ALLOW Anywhere

```

**Figure 3:** UFW firewall rule allowing TCP traffic on port 5001